BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 127 · Lesson 127 · Lossless and Lossy Compression

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is indirect addressing in a CPU instruction?

Q2. What is the binary result of 1 + 1 in a full adder with carry-in 0?

Q3. What is O(n) search complexity associated with?

Q4. Which memory is typically volatile: RAM or ROM?

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I Do · Read new knowledge

Key vocabulary

lossless — Original data can be reconstructed exactly.

lossy — Some original information is deliberately discarded.

redundancy — Repeated or predictable data.

compression ratio — Original size divided by compressed size (by this convention).

New learning

Find redundant data — Repeated patterns and predictable structures can be encoded efficiently.

Choose method — Lossless retains all detail; lossy permits irreversible approximation.

Measure impact — Consider file size, compression ratio, quality and purpose.

Worked example

A medical image must remain identical after decompression. Choose a technique and justify.

1. Medical image data — Predict the first step from the prompt.

2. Lossless encoding — Explain the intermediate result or decision.

3. Exact reconstruction — Lossless compression is required if exact original image data must be preserved. Lossy compression permanently discards information and could alter diagnostically important features.

We Do · Practise with guidance

A streaming service sends video over a limited-bandwidth connection. Is lossy compression always unacceptable?

Limited bandwidth

Predict the first step from the prompt.

Choose acceptable bitrate

Explain the intermediate result or decision.

Consider visual quality

Not necessarily. Perceptually lossy encoding can reduce bitrate enough for smooth playback, but visible artefacts may appear. Suitability depends on quality, latency and the application.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define lossy and lossless compression and give one application of each. [6]

Question 2

A file is reduced from 12 MB to 3 MB. Calculate the original-to-compressed compression ratio. [4]

Question 3

Evaluate a compression choice for source code, an online photograph and an MRI scan. [8]

Exit Ticket · Answer before checking

A lossy method always reconstructs the original bit-for-bit.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The instruction points to a location holding the effective address.

Q2. Sum 0, carry-out 1.

Q3. Linear/sequential search in the worst case.

Q4. RAM.

You Do pages · Independent questions

Q1. Lossless preserves exact data [2] with source code/archives [1]; lossy irreversibly discards detail [2] with streaming images/audio/video [1].

Q2. 12:3 reduces to 4:1 [4], or factor 4.

Q3. Source code lossless [2], photo possibly lossy [2], diagnostic scan lossless if exact data required [2], trade-off and qualified reasoning [2].

Exit Ticket · Model answer

False. Lossy compression irreversibly discards some information.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: